# Week 2

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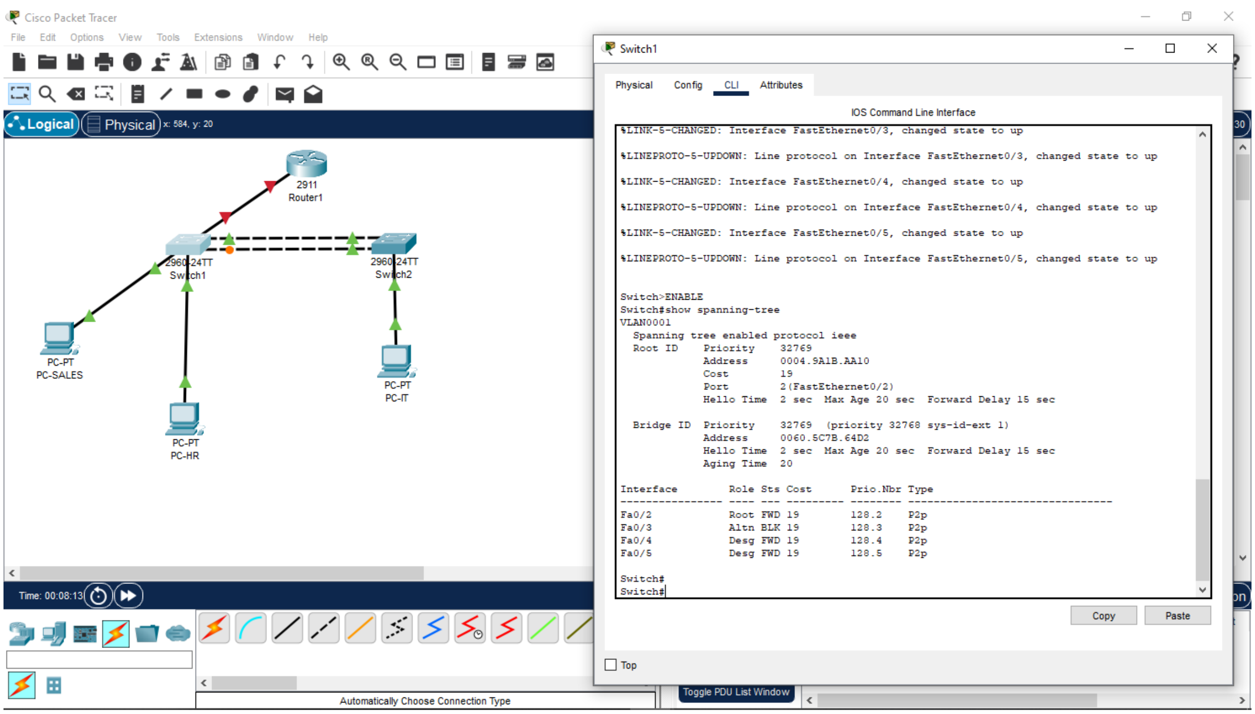
Week 2 practicals

**Tasks:**

**Step 1: Observe Default STP Behaviour**

1. show spanning-tree

2. Identify the root bridge and the port that STP has placed into blocking state.



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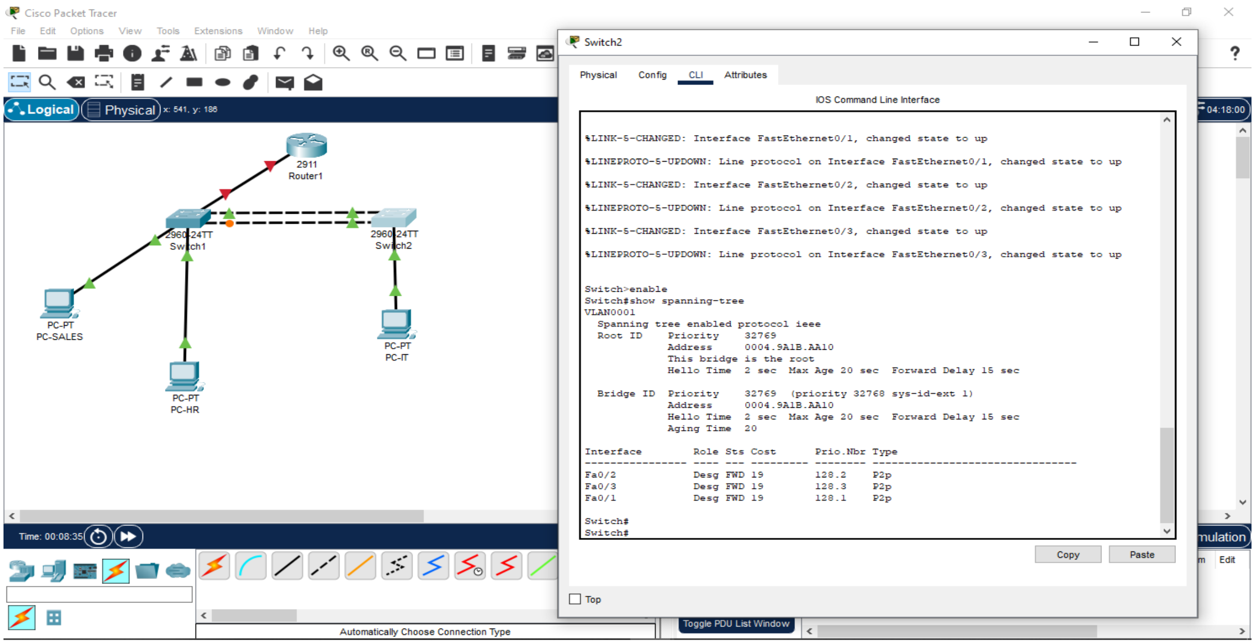
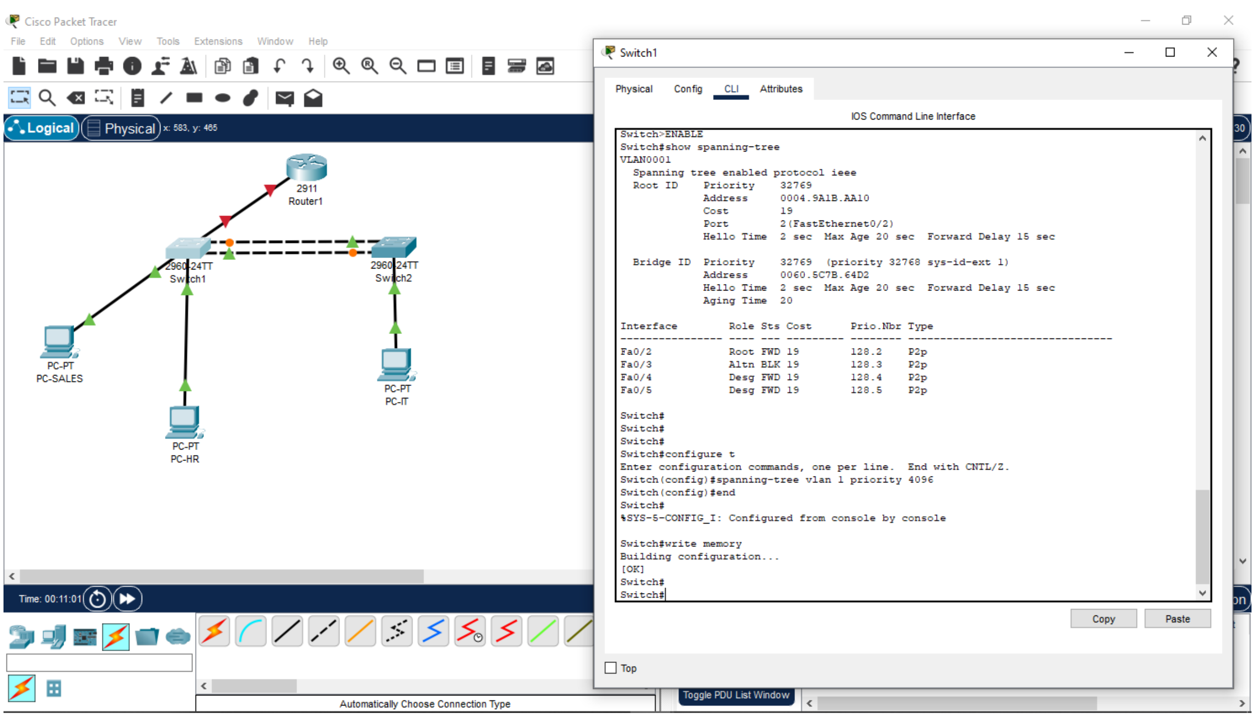



**Step 2: Tune STP**

1. On SW1: spanning-tree vlan 1 priority 4096 (to force SW1 as root bridge).

2. Shut down the currently active (forwarding) link between SW1 and SW2 and time how long it takes the blocked link to take over.

3. Re-enable the link once you've documented convergence.



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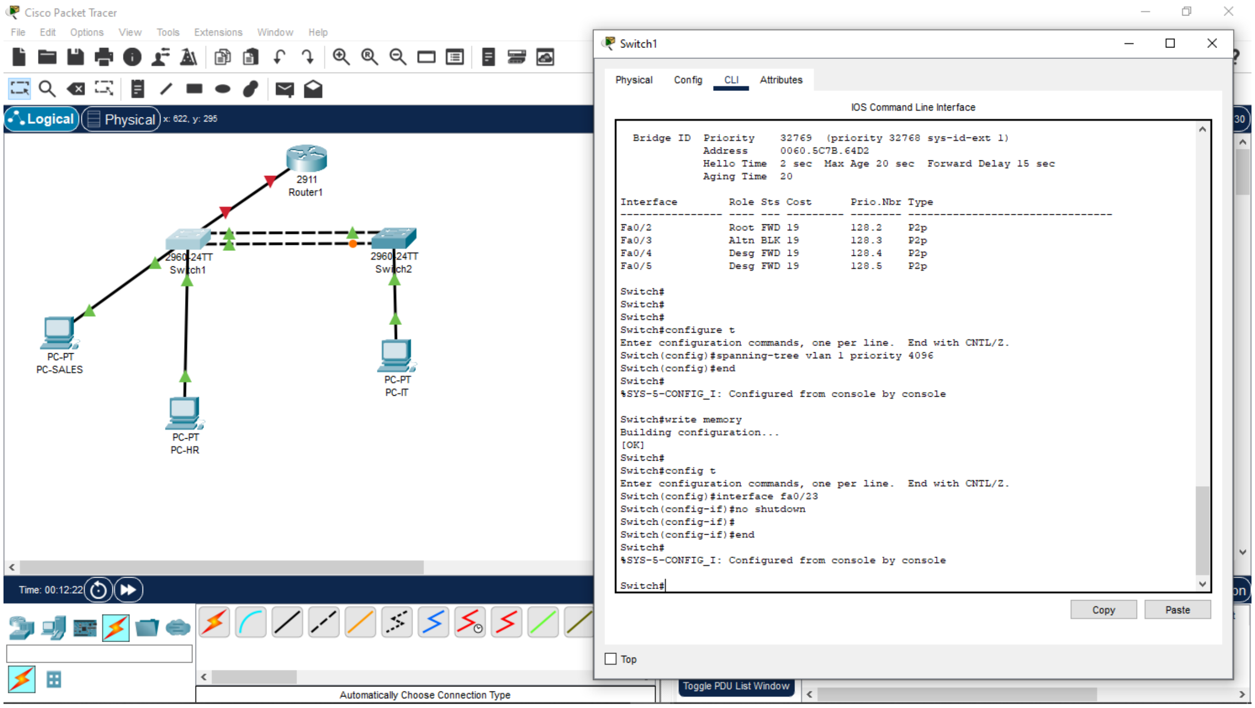
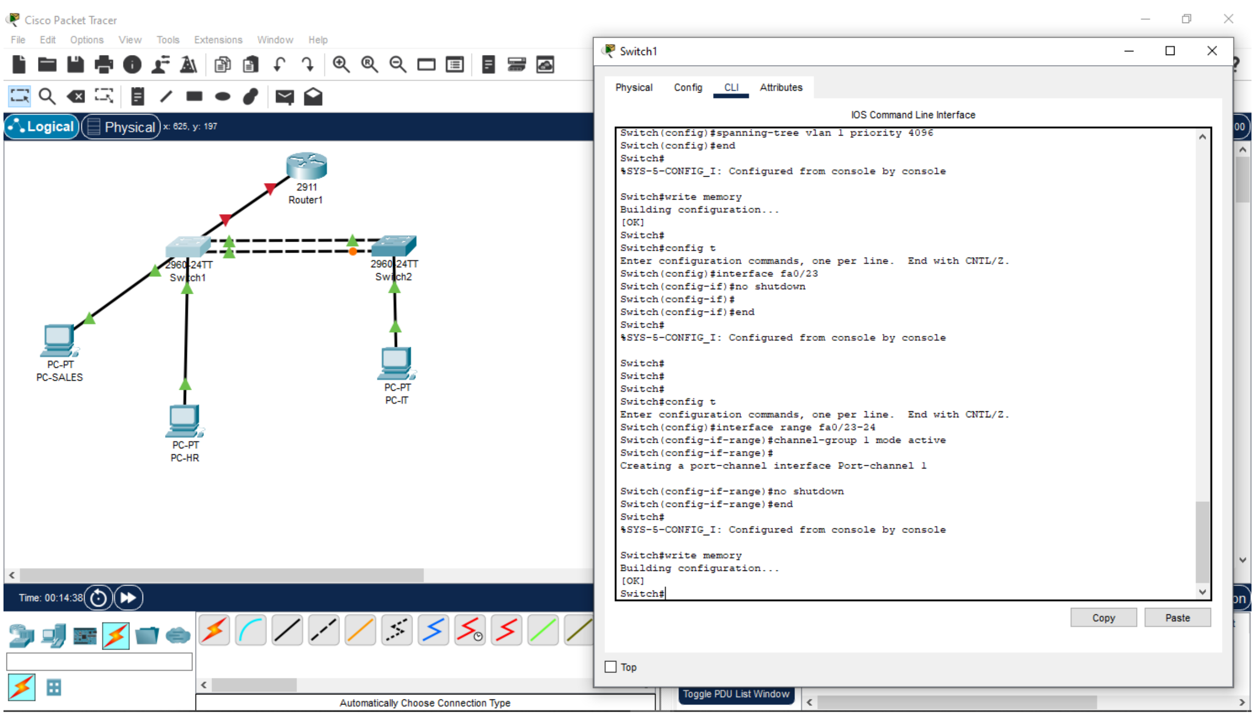



**Step 3: Replace the Redundant Links with EtherChannel**

1. On both switches: interface range fa0/23-24 channel-group 1 mode active

2. show etherchannel summary



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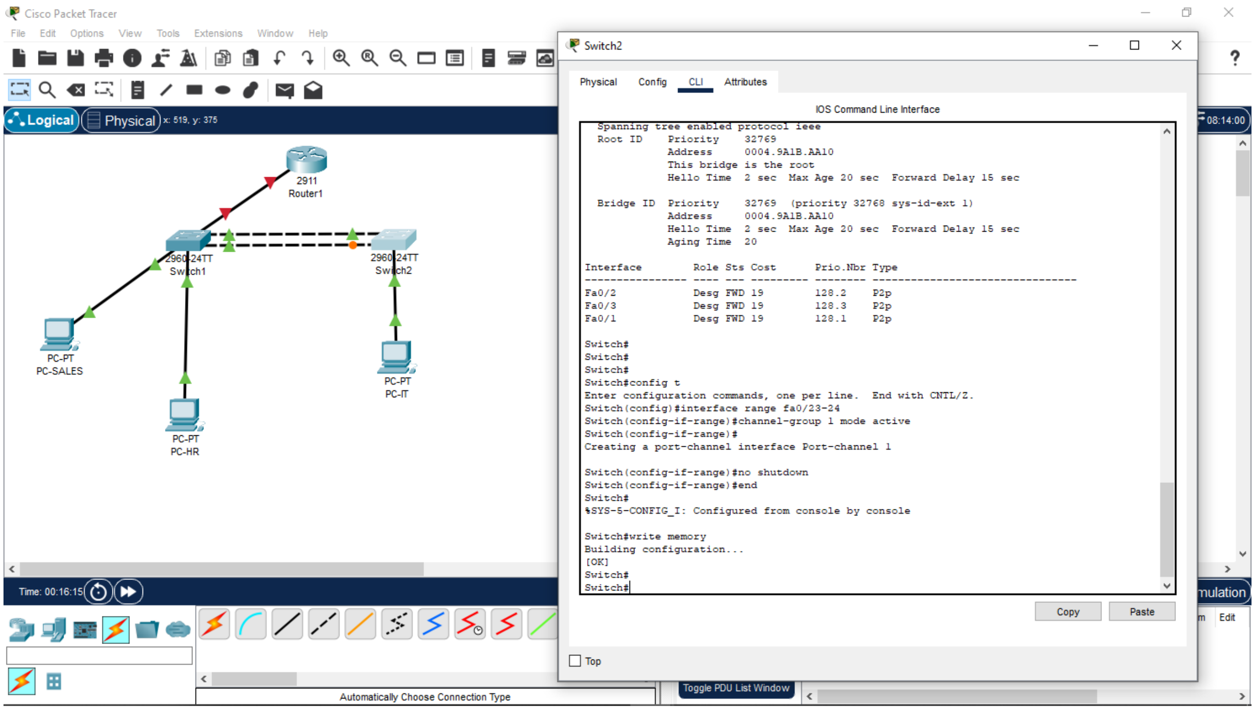
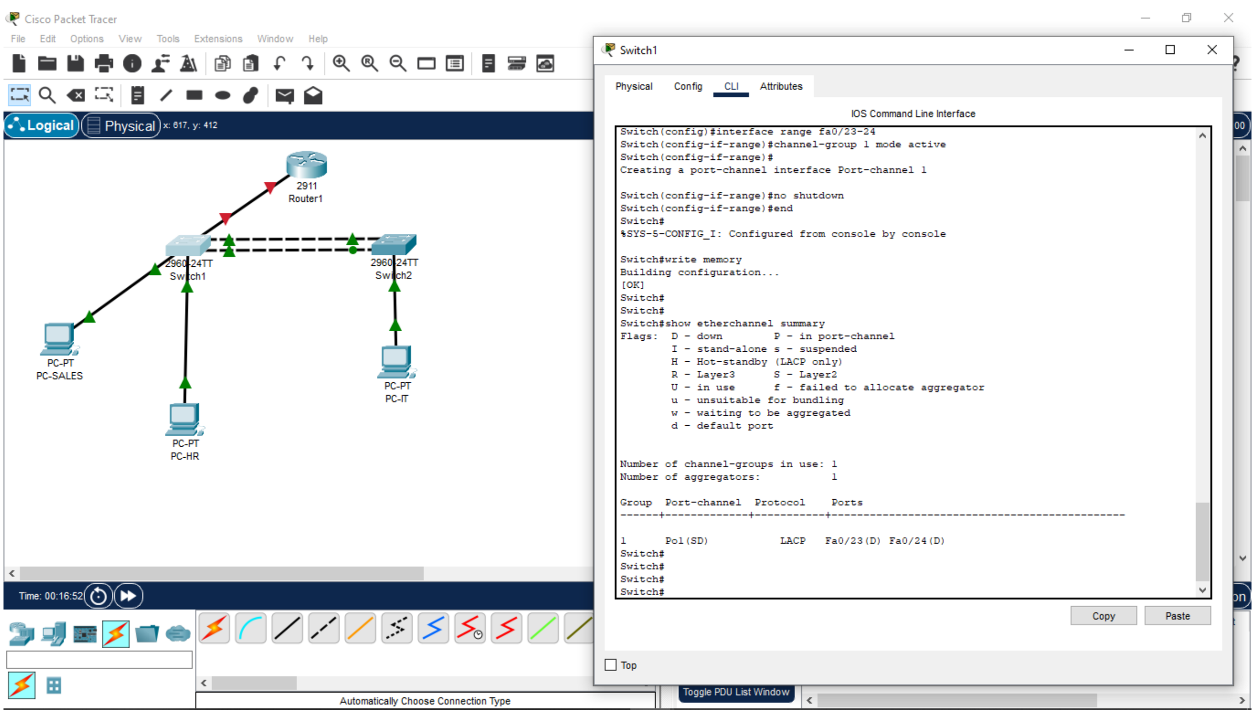


Cisco Packet Tracer
Switch2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0004.9A1B.AA10
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0004.9A1B.AA10
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/3 Desg FWD 19 128.3 P2p
Fa0/1 Desg FWD 19 128.1 P2p
Switch#
Switch#
Switch#
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/23-24
Switch(config-if-range)#channel-group 1 mode active
Switch(config-if-range)#
Creating a port-channel interface Port-channel 1
Switch(config-if-range)#no shutdown
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
Copy
Paste
Top
2911
Router1
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Logical
Physical
Time: 00:16:15
Automatically Choose Connection Type
Toggle PDU List Window

Cisco Packet Tracer
Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#interface range fa0/23-24
Switch(config-if-range)#channel-group 1 mode active
Switch(config-if-range)#
Creating a port-channel interface Port-channel 1
Switch(config-if-range)#no shutdown
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#
Switch#show etherchannel summary
Flags: D - down P - in port-channel
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3 S - Layer2
U - in use f - failed to allocate aggregator
u - unsuitable for bundling
w - waiting to be aggregated
d - default port
Number of channel-groups in use: 1
Number of aggregators: 1
Group Port-channel Protocol Ports
1 Po1(SD) LACP Fa0/23(D) Fa0/24(D)
Switch#
Switch#
Switch#
Switch#
Copy
Paste
Top
2911
Router1
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Logical
Physical
Time: 00:16:52
Automatically Choose Connection Type
Toggle PDU List Window

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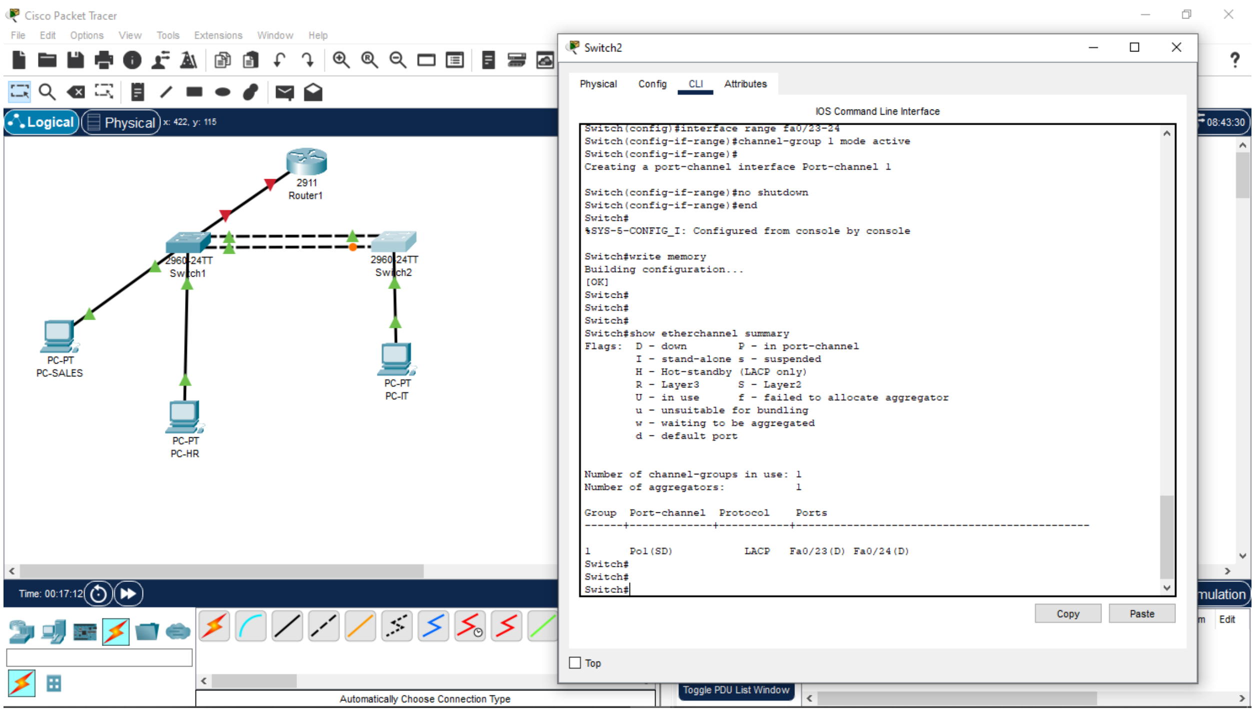


**Reflection Questions (answer these in your submission):**

1. What would happen to your VLAN traffic on this topology if STP were disabled entirely?

If STP is disabled with 2 physical links between SW1 and SW2, you would get a layer 2 loop.

The consequences are:

Broadcast, multicast, and unknown unicast frames loop forever between SW1 and SW2, consuming 100% bandwidth.

MAC Table Instability, switches keep seeing the same source MAC on different ports and keep updating the MAC table.

Network Meltdown, all VLAN traffic (VLAN10, VLAN20, VLAN30) would be impacted. PCs would lose connectivity.

That’s why we use STP to block redundant paths, and later EtherChannel to use both links safely as one.